1

For Task 1 I will change the equations into helper functions and show that these functions are continuous (because composing, adding etc of continuous functions results in continuous functions) and then I will show that the helper function has a sign change within a certain interval. Using the Intermediate Value Theorem I can therefore conclude that there is a root within that interval.

(a)

Define the function $h : \mathbb{R} \to \mathbb{R}$ by:

$$h(x) := \sum_{n=1}^{1234} \frac{x^n}{n^5} - \sqrt{2}.$$

Continuity: The function h is a polynomial in x (a finite sum of power functions), so it is continuous on $\mathbb{R}$.

Sign change: First, we evaluate h at $x = 0$:

$$h(0) = 0 - \sqrt{2} < 0.$$

Next, we evaluate h at $x = 2$:

$$h(2) = \sum_{n=1}^{1234} \frac{2^n}{n^5} - \sqrt{2}.$$

Considering just the first term of the sum ($n = 1$), we have $\frac{2^1}{1^5} = 2$. Since all terms for $x = 2$ are positive:

$$\sum_{n=1}^{1234} \frac{2^n}{n^5} > 2.$$

Since $\sqrt{2} < 2$, it follows that $h(2) > 2 - \sqrt{2} > 0$.

Conclusion: Since h is continuous and changes sign on the interval $[0, 2]$, by IVT there exists at least one $x_0 \in (0, 2)$ such that $h(x_0) = 0$.

(b)

Define $h : \mathbb{R} \to \mathbb{R}$ by:

$$h(x) := x^{47} + x^4 - \sqrt{|x + 4|} - \sqrt{|x + 3|} + 3.$$

Continuity: The function is composed of continuous standard functions (polynomials, absolute values, and square roots). Since $|x + 4|$ and $|x + 3|$ are non-negative for all x, the square roots are defined on $\mathbb{R}$. Thus, h is continuous on $\mathbb{R}$.

Sign change: First, consider $x = 0$:

$$\begin{aligned} h(0) &= 0^{47} + 0^4 - \sqrt{|0+4|} - \sqrt{|0+3|} + 3 \\ &= -\sqrt{4} - \sqrt{3} + 3 \\ &= -2 - \sqrt{3} + 3 \\ &= 1 - \sqrt{3}. \end{aligned}$$

Since $1^2 = 1$ and $(\sqrt{3})^2 = 3$, and $1 < 3$, it follows that $1 < \sqrt{3}$. Therefore, $h(0) < 0$.

Next, consider $x = 1$:

$$\begin{aligned} h(1) &= 1^{47} + 1^4 - \sqrt{|1+4|} - \sqrt{|1+3|} + 3 \\ &= 1 + 1 - \sqrt{5} - \sqrt{4} + 3 \\ &= 5 - \sqrt{5} - 2 \\ &= 3 - \sqrt{5}. \end{aligned}$$

To determine the sign, we compare 3 and $\sqrt{5}$ by squaring both values:

$$3^2 = 9 \quad \text{and} \quad (\sqrt{5})^2 = 5.$$

Since $9 > 5$, it implies $3 > \sqrt{5}$. Therefore, $h(1) > 0$.

Conclusion: Since h is continuous on $[0, 1]$ and $h(0) < 0 < h(1)$, the IVT implies there exists some $x_0 \in (0, 1)$ such that $h(x_0) = 0$.

(c)

We define the function $h : \mathbb{R} \to \mathbb{R}$ as:

$$h(x) := f(x) - g(x^6 - 1).$$

Since f and g are continuous functions and $x \mapsto x^6 - 1$ is a polynomial (continuous), the composition h is continuous on $\mathbb{R}$.

We evaluate h at the points $x = 1$ and $x = -1$ using the given values ($f(1) = 3, g(0) = 2, f(-1) < 0$):

1. **At $x = 1$:**

$$\begin{aligned} h(1) &= f(1) - g(1^6 - 1) \\ &= f(1) - g(0) \\ &= 3 - 2 = 1. \end{aligned}$$

Thus, $h(1) > 0$.

2. **At $x = -1$:**

$$\begin{aligned} h(-1) &= f(-1) - g((-1)^6 - 1) \\ &= f(-1) - g(0) \\ &= f(-1) - 2. \end{aligned}$$

Since it is given that $f(-1) < 0$, it follows that $f(-1) - 2 < -2$. Thus, $h(-1) < 0$.

Conclusion: Since h is continuous on the interval $[-1, 1]$ and changes sign ($h(-1) < 0 < h(1)$), the IVT implies that there exists at least one $x_0 \in (-1, 1)$ such that $h(x_0) = 0$.

(d)

We rearrange the equation to define the helper function $F : \mathbb{R} \to \mathbb{R}$:

$$F(x) := h(x) + h(x)^{27}.$$

Continuity: Since h is continuous (given) and the power function $y \mapsto y^{27}$ is continuous, their composition and sum F is continuous on $\mathbb{R}$.

Sign Change: The problem states that $h(\mathbb{R}) = \mathbb{R}$, which means h is surjective. It takes on every real value.

1. There exists some $x_1 \in \mathbb{R}$ such that $h(x_1) = 1$. Evaluating F at this point:

$$F(x_1) = 1 + 1^{27} = 2 > 0.$$

2. There exists some $x_2 \in \mathbb{R}$ such that $h(x_2) = -1$. Evaluating F at this point:

$$F(x_2) = -1 + (-1)^{27} = -1 - 1 = -2 < 0.$$

Conclusion: Since F is continuous and takes values with opposite signs ($F(x_2) < 0 < F(x_1)$), the IVT implies that there exists an x_0 between x_1 and x_2 such that $F(x_0) = 0$.

2

(a)

Let $x, y \in [2019, 2020]$. Let's estimate the difference $|a(x) - a(y)|$:

$$\begin{aligned}
|a(x) - a(y)| &= |(x+1)^2 - (y+1)^2| \\
&= |((x+1) - (y+1)) \cdot ((x+1) + (y+1))| \quad \text{(Difference of squares)} \\
&= |(x-y) \cdot (x+y+2)| \\
&= |x-y| \cdot |x+y+2|.
\end{aligned}$$

Since x and y are in the interval $[2019, 2020]$, we can bound the second factor:

$$|x+y+2| \leq |x| + |y| + 2 \leq 2020 + 2020 + 2 = 4042.$$

Thus, for all $x, y \in [2019, 2020]$, we have:

$$|a(x) - a(y)| \leq 4042|x-y|.$$

Conclusion: Let $\varepsilon > 0$. We choose $\delta := \frac{\varepsilon}{4042}$. Then for all $x, y \in [2019, 2020]$ with $|x - y| < \delta$, it holds that:

$$|a(x) - a(y)| \leq 4042|x-y| < 4042 \cdot \frac{\varepsilon}{4042} = \varepsilon.$$

Therefore, a is uniformly continuous on $[2019, 2020]$.

Solution to 2(b) using Script Tools

We check the uniform continuity of $b(x) = \sqrt[4]{x}$ on $D = [0, \infty)$. We split the domain into $D_1 = [0, 1]$ and $D_2 = [1, \infty)$.

1. On the interval $D_1 = [0, 1]$: According to **Theorem 4.4.8**, the inverse function of the continuous, strictly monotone function $x \mapsto x^4$ is continuous. Thus, $b(x) = \sqrt[4]{x}$ is continuous on D_1. Since D_1 is a compact interval (closed and bounded), the function is uniformly continuous on D_1 (Heine-Cantor Theorem).

2. On the interval $D_2 = [1, \infty)$: We use the algebraic expansion for differences of powers from **Example 4.4.9**. Let $u = \sqrt[4]{x}$ and $v = \sqrt[4]{y}$. Then $x = u^4$ and $y = v^4$, and:

$$|x - y| = |u^4 - v^4| = |u - v| \cdot (u^3 + u^2v + uv^2 + v^3).$$

Rearranging for $|b(x) - b(y)| = |u - v|$:

$$|b(x) - b(y)| = \frac{|x - y|}{u^3 + u^2v + uv^2 + v^3}.$$

For $x, y \in [1, \infty)$, we have $u \geq 1$ and $v \geq 1$. Thus, the denominator can be estimated as:

$$u^3 + u^2v + uv^2 + v^3 \geq 1 + 1 + 1 + 1 = 4.$$

This yields the Lipschitz condition:

$$|b(x) - b(y)| \leq \frac{1}{4}|x - y|.$$

Since the function is Lipschitz continuous on D_2, it is uniformly continuous on D_2.

Conclusion: Since b is uniformly continuous on $[0, 1]$ and $[1, \infty)$, it is uniformly continuous on the union $[0, \infty)$.

(d)

Let $x, y \in (5, \infty)$. We estimate the difference:

$$\begin{aligned} |d(x) - d(y)| &= \left| \frac{1}{x^2} - \frac{1}{y^2} \right| \\ &= \left| \frac{y^2 - x^2}{x^2y^2} \right| \\ &= |x - y| \cdot \frac{x + y}{x^2y^2}. \end{aligned}$$

To bound the second factor, we use the fact that $x > 5$ and $y > 5$. We rewrite the term:

$$\frac{x + y}{x^2y^2} = \frac{x}{x^2y^2} + \frac{y}{x^2y^2} = \frac{1}{xy^2} + \frac{1}{x^2y}.$$

Using the lower bounds of the domain:

$$\frac{1}{xy^2} < \frac{1}{5 \cdot 5^2} = \frac{1}{125} \quad \text{and} \quad \frac{1}{x^2y} < \frac{1}{5^2 \cdot 5} = \frac{1}{125}.$$

Adding these estimates:

$$\frac{x+y}{x^2y^2} < \frac{1}{125} + \frac{1}{125} = \frac{2}{125}.$$

Thus, for all $x, y \in (5, \infty)$:

$$|d(x) - d(y)| \leq \frac{2}{125}|x - y|.$$

Conclusion: Let $\varepsilon > 0$. We choose $\delta := \frac{125}{2}\varepsilon$. Then for all $x, y \in (5, \infty)$ with $|x - y| < \delta$:

$$|d(x) - d(y)| \leq \frac{2}{125}|x - y| < \frac{2}{125} \cdot \left(\frac{125}{2}\varepsilon\right) = \varepsilon.$$

Therefore, d is uniformly continuous on $(5, \infty)$.

3

(a) Proof: Strict Monotonicity

Assumption: Assume f is *not* strictly monotone.

If f is not strictly monotone, there exist three points $a, b, c \in I$ with $a < b < c$ such that $f(b)$ does not lie between $f(a)$ and $f(c)$. This implies either a local maximum or a local minimum occurs between a and c.

Without loss of generality, assume a local maximum:

$$f(b) > f(a) \quad \text{and} \quad f(b) > f(c).$$

Let η be a value strictly between the peak and the larger of the endpoints:

$$\max(f(a), f(c)) < \eta < f(b).$$

Since f is continuous, we apply the IVT:

1. On the interval $[a, b]$, since $f(a) < \eta < f(b)$, there exists $x_1 \in (a, b)$ such that $f(x_1) = \eta$.
2. On the interval $[b, c]$, since $f(c) < \eta < f(b)$, there exists $x_2 \in (b, c)$ such that $f(x_2) = \eta$.

Since $x_1 < b < x_2$, we have found two distinct points $x_1 \neq x_2$ such that $f(x_1) = f(x_2) = \eta$.

This contradicts the injectivity of f (which is required for the inverse f^{-1} to exist). Therefore, our assumption is false, and f must be strictly monotone.

(b) Proof: Properties of f with a Jump

Let f be strictly monotonically increasing (WLOG) with exactly one jump discontinuity at $x_0 \in \mathring{I}$.

1. Existence of f^{-1}

Since f is strictly monotone, it is injective. By definition, f maps I surjectively onto its image $J := f(I)$. Thus, $f : I \to J$ is a bijection, and the inverse function $f^{-1} : J \to I$ exists.

2. Strict Monotonicity of f^{-1}

Consider $y_1, y_2 \in J$ with $y_1 < y_2$. Let $x_1 = f^{-1}(y_1)$ and $x_2 = f^{-1}(y_2)$. Assume for contradiction that $x_1 \geq x_2$. Since f is increasing, this implies $f(x_1) \geq f(x_2) \implies y_1 \geq y_2$, which contradicts the premise $y_1 < y_2$. Thus, $x_1 < x_2$, proving f^{-1} is strictly monotonically increasing.

3. J is not an interval

Since f has a jump at x_0, the one-sided limits exist but are distinct (Def 4.4.4):

$$\alpha := \lim_{x \to x_0^-} f(x) < \lim_{x \to x_0^+} f(x) =: \beta.$$

Because f is strictly increasing:

- For all $x < x_0$, $f(x) \leq \alpha$.
- For all $x > x_0$, $f(x) \geq \beta$.

The open interval (α, β) contains no points of J (except possibly the single value $f(x_0)$). Since $\alpha < \beta$, there is a "gap" in the values of f. Therefore, the set J is disconnected and cannot be an interval.

4. Continuity of f^{-1}

We know that $f^{-1} : J \to I$ is strictly monotone and its image is I. Since I is an interval (given), it contains no gaps. A monotone function can only have jump discontinuities. If f^{-1} had a jump discontinuity at some $y_0 \in J$, the image of f^{-1} would skip an interval of values in I. However, the image is exactly the connected interval I. Therefore, f^{-1} cannot have any jumps and must be continuous.

4

(a)

We apply the addition theorems (Theorem 4.5.3) for $w = \pi$. Using the known values $\cos(\pi) = -1$ and $\sin(\pi) = 0$:

$$\begin{aligned}\cos(z+\pi) &= \cos(z)\cos(\pi) - \sin(z)\sin(\pi)\\ &= \cos(z)\cdot(-1) - \sin(z)\cdot 0\\ &= -\cos(z).\end{aligned}$$

$$\begin{aligned}\sin(z+\pi) &= \sin(z)\cos(\pi) + \cos(z)\sin(\pi)\\ &= \sin(z)\cdot(-1) + \cos(z)\cdot 0\\ &= -\sin(z).\end{aligned}$$

(b)

We apply the addition theorems for $w = 2k\pi$ with $k \in \mathbb{Z}$. From Theorem 4.5.12(1), we know that $e^{2k\pi i} = 1$. Comparing real and imaginary parts implies:

$$\cos(2k\pi) = 1 \quad \text{and} \quad \sin(2k\pi) = 0.$$

Substituting these values:

$$\begin{aligned}\cos(z+2k\pi) &= \cos(z)\cos(2k\pi) - \sin(z)\sin(2k\pi)\\ &= \cos(z)\cdot 1 - \sin(z)\cdot 0\\ &= \cos(z).\end{aligned}$$

$$\begin{aligned}\sin(z+2k\pi) &= \sin(z)\cos(2k\pi) + \cos(z)\sin(2k\pi)\\ &= \sin(z)\cdot 1 + \cos(z)\cdot 0\\ &= \sin(z).\end{aligned}$$

(c)

1. Zeros of the Sine Function We start with the definition of the sine function (Theorem 4.5.2 (7)):

$$\sin(z) = \frac{e^{iz} - e^{-iz}}{2i}.$$

We set the expression to zero and solve for z:

$$\begin{aligned}\sin(z) = 0 &\iff \frac{e^{iz} - e^{-iz}}{2i} = 0\\ &\iff e^{iz} - e^{-iz} = 0\\ &\iff e^{iz} = e^{-iz}.\end{aligned}$$

Since $e^{iz} \neq 0$ for all $z \in \mathbb{C}$ (Theorem 4.5.2 (4)), we can multiply both sides of the equation by e^{iz}:

$$e^{iz} \cdot e^{iz} = e^{-iz} \cdot e^{iz} \implies e^{2iz} = e^0 = 1.$$

According to Theorem 4.5.12 (1), the exponential function $e^w = 1$ if and only if $w \in 2\pi i\mathbb{Z}$. Substituting $w = 2iz$:

$$2iz = 2k\pi i \quad \text{for some } k \in \mathbb{Z}.$$

Dividing by $2i$, we obtain:

$$z = k\pi.$$

Thus, the set of zeros is $\{k\pi \mid k \in \mathbb{Z}\} = \pi\mathbb{Z}$.

2. Zeros of the Cosine Function We use the definition of the cosine function:

$$\cos(z) = \frac{e^{iz} + e^{-iz}}{2}.$$

Setting this to zero:

$$\begin{aligned} \cos(z) = 0 &\iff \frac{e^{iz} + e^{-iz}}{2} = 0 \\ &\iff e^{iz} = -e^{-iz}. \end{aligned}$$

Multiplying both sides by e^{iz}:

$$e^{2iz} = -1.$$

According to Theorem 4.5.12 (2), the exponential function $e^w = -1$ if and only if w is an odd multiple of πi, i.e., $w \in \pi i + 2\pi i\mathbb{Z}$. Substituting $w = 2iz$:

$$2iz = (2k+1)\pi i \quad \text{for some } k \in \mathbb{Z}.$$

Dividing by $2i$:

$$z = \frac{(2k+1)\pi}{2} = \frac{\pi}{2} + k\pi.$$

Thus, the set of zeros is $\{\frac{\pi}{2} + k\pi \mid k \in \mathbb{Z}\} = \frac{\pi}{2} + \pi\mathbb{Z}$.

(d)

1. Boundedness ($\subseteq [-1, 1]$) Let $x \in \mathbb{R}$. According to Theorem 4.5.2 (2), $\cos(x)$ and $\sin(x)$ are real-valued. Using the trigonometric identity from Theorem 4.5.3 (3):

$$\cos^2(x) + \sin^2(x) = 1.$$

Since squares of real numbers are non-negative ($\cos^2(x) \geq 0$ and $\sin^2(x) \geq 0$), it follows that:

$$\cos^2(x) \leq 1 \quad \text{and} \quad \sin^2(x) \leq 1.$$

Taking the square root implies $|\cos(x)| \leq 1$ and $|\sin(x)| \leq 1$. Thus, $\cos(\mathbb{R}) \subseteq [-1, 1]$ and $\sin(\mathbb{R}) \subseteq [-1, 1]$.

2. Surjectivity ($\supseteq [-1, 1]$) We prove that the cosine function takes every value in $[-1, 1]$ using the Intermediate Value Theorem.

- **Continuity:** According to Theorem 4.5.2 (9), the function $\cos : \mathbb{R} \to \mathbb{R}$ is continuous.
- **Values at Endpoints:**
  - At $x = 0$: $\cos(0) = \mathrm{Re}(e^{i0}) = 1$ (Theorem 4.5.4(1)).
  - At $x = \pi$: From Theorem 4.5.12(2), we know $e^{i\pi} = -1$. Thus, $\cos(\pi) = \mathrm{Re}(-1) = -1$.
- **Application of IVT:** Since cos is continuous on the interval $[0, \pi]$ and takes the values 1 and -1, by the Intermediate Value Theorem (Theorem 4.4.7), it must take every value y between -1 and 1.

$$[-1, 1] \subseteq \cos([0, \pi]) \subseteq \cos(\mathbb{R}).$$

Therefore, $\cos(\mathbb{R}) = [-1, 1]$.

For the sine function, we use the shift identity from Theorem 4.5.14 (5):

$$\sin(x) = \cos\left(\frac{\pi}{2} - x\right).$$

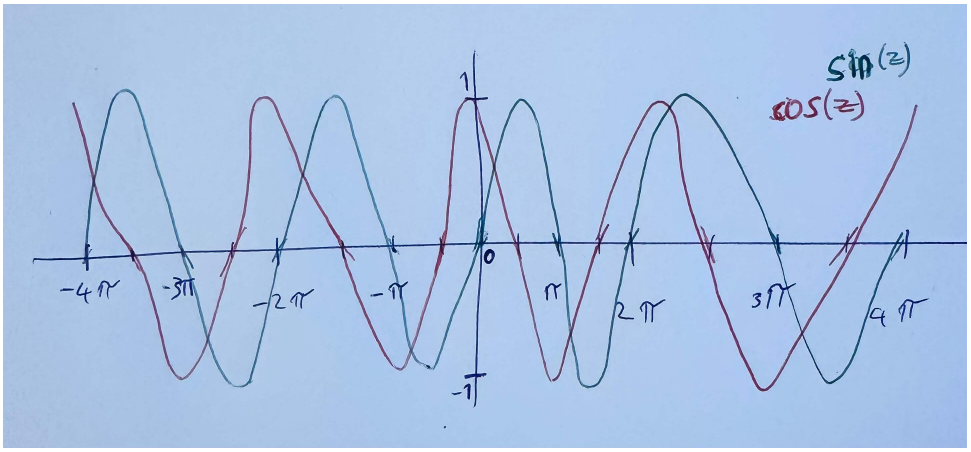
Since the map $x \mapsto \frac{\pi}{2} - x$ is a bijection from $\mathbb{R}$ onto $\mathbb{R}$, the image of the sine function is exactly the same as the image of the cosine function.

$$\sin(\mathbb{R}) = \cos(\mathbb{R}) = [-1, 1].$$

Graph

Sketching the functions on the interval $[-4\pi, 4\pi]$:

- **Periodicity:** Both functions are 2π-periodic (Theorem 4.5.14(1)). The interval $[-4\pi, 4\pi]$ covers exactly 2 periods to the left and 2 periods to the right of the origin.
- **Zeros:**
  - $\sin(x) = 0$ at integers of π $(0, \pm\pi, \pm 2\pi, \dots)$.
  - $\cos(x) = 0$ at half-integers of π $(\pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \dots)$.
- **Extrema:**
  - Maxima $(+1)$ and Minima (-1) alternate every π.
  - $\cos(x)$ starts at 1 at $x = 0$.
  - $\sin(x)$ starts at 0 at $x = 0$.


sin(z)
cos(z)
1
-1
0
-4π
-3π
-2π
-π
π
2π
3π
4π